AI, ML & MLOps Interview Preparation Quiz – Answer Sheet

Birla Prabhu K

Score: **180 / 250** (72.0%) · Grade: **B** · Rank: **16 of 29** · Correct: 180 · Wrong: 70 · Not answered: 0

Grade bands: A+ 90%+ · A 80–89% · B 70–79% · C 60–69% · D 50–59% · below 50% Needs improvement

Topic-wise score

Topic	Score	%
Machine Learning Fundamentals	22 / 22	100%
Deep Learning	17 / 18	94%
Large Language Models (LLMs)	20 / 23	87%
Prompt Engineering	14 / 15	93%
Retrieval-Augmented Generation (RAG)	18 / 20	90%
Advanced & Multi-RAG	10 / 10	100%
Harness Engineering & AI Agents	10 / 10	100%
Model Context Protocol (MCP) & MCP Inspector	7 / 20	35%
Ollama & Local LLMs	2 / 12	17%
MLOps & MLflow	7 / 20	35%
Streamlit	3 / 10	30%
Docker	4 / 15	27%
Kubernetes (K8s)	6 / 15	40%
Git & GitHub	12 / 12	100%
APIs & FastAPI	12 / 12	100%
Deployment (Render)	6 / 6	100%
Jenkins & CI/CD	10 / 10	100%

Topics shaded red are below 60% – revise these first.

Question-by-question review (✓ = correct, ✗ = wrong, – = not answered)

#	Question	Your answer	Correct answer		Explanation
Machine Learning Fundamentals					
1	Supervised learning requires which type of data?	D) Labeled training data	D) Labeled training data	✓	Supervised learning learns a mapping from inputs to known labels/targets.
2	Which of these is a classification algorithm?	A) Logistic Regression	A) Logistic Regression	✓	Despite its name, logistic regression predicts class probabilities.
3	K-Means is best described as:	A) An unsupervised clustering algorithm	A) An unsupervised clustering algorithm	✓	K-Means groups unlabeled points into k clusters by distance to centroids.
4	A model performs very well on training data but poorly on test data. This is:	C) Overfitting	C) Overfitting	✓	Overfitting means the model memorized noise in training data and fails to generalize.
5	High bias in a model typically leads to:	D) Underfitting	D) Underfitting	✓	High bias = overly simple assumptions, so the model misses real patterns.
6	Which technique helps reduce overfitting?	C) L1/L2 regularization	C) L1/L2 regularization	✓	Regularization penalizes large weights, discouraging overly complex models.
7	L1 regularization (Lasso) tends to:	C) Drive some weights to exactly zero, performing feature selection	C) Drive some weights to exactly zero, performing feature selection	✓	The L1 penalty produces sparse solutions, effectively dropping features.
8	Precision is calculated as:	A) TP / (TP + FP)	A) TP / (TP + FP)	✓	Precision = of everything predicted positive, how many were actually positive.
9	Recall (sensitivity) is calculated as:	B) TP / (TP + FN)	B) TP / (TP + FN)	✓	Recall = of all actual positives, how many the model found.
10	For a fraud dataset where only 1% of transactions are fraud, which metric is most useful?	C) F1-score or PR-AUC	C) F1-score or PR-AUC	✓	Accuracy is misleading on imbalanced data: predicting 'not fraud' always gives 99%.
11	The F1-score is:	B) The harmonic mean of precision and recall	B) The harmonic mean of precision and recall	✓	F1 = 2PR/(P+R); it is high only when both precision and recall are high.

#	Question	Your answer	Correct answer		Explanation
12	What is the main purpose of k-fold cross-validation?	C) To estimate model performance on unseen data more reliably	C) To estimate model performance on unseen data more reliably	✓	Each fold serves once as validation, giving a more stable performance estimate.
13	PCA (Principal Component Analysis) is mainly used for:	A) Dimensionality reduction	A) Dimensionality reduction	✓	PCA projects data onto directions of maximum variance, reducing feature count.
14	Data leakage in ML refers to:	D) Information from outside the training data (e.g., the target or test set) influencing training, inflating performance	D) Information from outside the training data (e.g., the target or test set) influencing training, inflating performance	✓	Leakage gives unrealistically good validation scores that collapse in production.
15	Gradient descent works by:	D) Iteratively updating parameters in the direction of the negative gradient to minimize loss	D) Iteratively updating parameters in the direction of the negative gradient to minimize loss	✓	Moving opposite to the gradient reduces the loss step by step.
16	If the learning rate is set too high, what typically happens?	A) The loss oscillates or diverges	A) The loss oscillates or diverges	✓	Steps that are too large overshoot the minimum.
17	Random Forest is:	A) An ensemble of decision trees trained with bagging	A) An ensemble of decision trees trained with bagging	✓	Random Forest averages many trees built on bootstrap samples with random feature subsets.
18	XGBoost is based on which technique?	D) Gradient boosting	D) Gradient boosting	✓	XGBoost builds trees sequentially, each correcting errors of the previous ones.
19	Feature scaling (standardization) is most important for:	C) Distance- or gradient-based algorithms like KNN, SVM and neural networks	C) Distance- or gradient-based algorithms like KNN, SVM and neural networks	✓	Unscaled features with large ranges dominate distances and slow gradient descent.
20	One-hot encoding is used to:	B) Convert categorical variables into binary vectors	B) Convert categorical variables into binary vectors	✓	Each category becomes its own 0/1 column.
21	ROC-AUC measures:	C) How well a model ranks positives above negatives across all thresholds	C) How well a model ranks positives above negatives across all thresholds	✓	AUC of 0.5 is random ranking; 1.0 is perfect separation.
22	The bias-variance tradeoff states that:	A) Reducing bias often increases variance and vice versa	A) Reducing bias often increases variance and vice versa	✓	Model complexity balances underfitting (bias) against overfitting (variance).
Deep Learning					
23	Why do neural networks use activation functions?	A) To introduce non-linearity	A) To introduce non-linearity	✓	Without non-linear activations, stacked layers collapse into one linear transformation.
24	The ReLU activation function is defined as:	A) max(0, x)	A) max(0, x)	✓	ReLU outputs x if positive, else 0.
25	The vanishing gradient problem is most associated with:	B) Sigmoid/tanh activations in very deep networks	B) Sigmoid/tanh activations in very deep networks	✓	Sigmoid/tanh derivatives are small, so gradients shrink as they are multiplied back through layers.
26	Backpropagation is used to:	B) Compute gradients of the loss with respect to weights using the chain rule	B) Compute gradients of the loss with respect to weights using the chain rule	✓	Backprop propagates error from output to input layers to compute gradients.
27	Dropout is a technique that:	D) Randomly deactivates neurons during training to reduce overfitting	D) Randomly deactivates neurons during training to reduce overfitting	✓	Dropout forces the network not to rely on any single neuron; it is disabled at inference.
28	Batch Normalization helps by:	A) Removing outliers from the dataset	B) Normalizing layer inputs over a mini-batch to stabilize and speed up training	✗	BatchNorm keeps activations in a stable range, allowing higher learning rates.
29	Convolutional Neural Networks (CNNs) are primarily used for:	A) Image and grid-like data	A) Image and grid-like data	✓	Convolutions capture local spatial patterns such as edges and textures.
30	What does a pooling layer in a CNN do?	A) Downsamples feature maps to reduce spatial size	A) Downsamples feature maps to reduce spatial size	✓	Max/average pooling reduces size and computation while keeping key features.
31	LSTM networks were introduced mainly to address:	B) Vanishing gradients and long-term dependencies in RNNs	B) Vanishing gradients and long-term dependencies in RNNs	✓	LSTM gates control information flow, letting it remember over long sequences.

#	Question	Your answer	Correct answer		Explanation
32	The key mechanism of the Transformer architecture is:	A) Self-attention	A) Self-attention	✓	Self-attention lets every token weigh every other token in parallel ('Attention Is All You Need').
33	The softmax function outputs:	D) A probability distribution that sums to 1	D) A probability distribution that sums to 1	✓	Softmax converts logits into class probabilities.
34	Cross-entropy loss is typically used for:	C) Classification tasks	C) Classification tasks	✓	It measures the gap between predicted probabilities and the true class.
35	One epoch in training means:	C) One complete pass over the entire training dataset	C) One complete pass over the entire training dataset	✓	Number of steps per epoch = dataset size / batch size.
36	Transfer learning means:	B) Reusing a model pretrained on one task as a starting point for a new task	B) Reusing a model pretrained on one task as a starting point for a new task	✓	E.g., fine-tuning an ImageNet-trained ResNet on medical images.
37	The Adam optimizer:	A) Combines momentum with adaptive per-parameter learning rates	A) Combines momentum with adaptive per-parameter learning rates	✓	Adam = momentum + RMSProp-style adaptive scaling.
38	Early stopping means:	B) Stopping training when validation performance stops improving	B) Stopping training when validation performance stops improving	✓	It prevents overfitting by halting before validation loss rises.
39	An embedding layer:	D) Maps discrete tokens/IDs to dense, learned vectors	D) Maps discrete tokens/IDs to dense, learned vectors	✓	Embeddings place similar items close together in vector space.
40	Why are GPUs preferred for deep learning?	D) They perform massively parallel matrix operations	D) They perform massively parallel matrix operations	✓	Neural network training is dominated by matrix multiplications that parallelize well.
Large Language Models (LLMs)					
41	What does LLM stand for?	C) Large Language Model	C) Large Language Model	✓	LLMs are transformer models trained on huge text corpora.
42	Tokenization in LLMs means:	A) Splitting text into tokens (often subwords) the model can process	A) Splitting text into tokens (often subwords) the model can process	✓	Methods like BPE break words into subword units mapped to IDs.
43	The 'context window' of an LLM is:	C) The maximum number of tokens the model can consider in one request (input + output)	C) The maximum number of tokens the model can consider in one request (input + output)	✓	Anything beyond the context window cannot be seen by the model.
44	The temperature parameter controls:	A) The model's size	B) The randomness of the generated output	✗	Lower temperature = more focused; higher = more varied/creative.
45	Setting temperature close to 0 makes outputs:	B) More deterministic, favoring the most likely tokens	B) More deterministic, favoring the most likely tokens	✓	Near-zero temperature is good for factual or structured tasks.
46	Top-p (nucleus) sampling:	C) Samples from the smallest set of tokens whose cumulative probability reaches p	C) Samples from the smallest set of tokens whose cumulative probability reaches p	✓	Top-p adapts the candidate pool size to the model's confidence.
47	An LLM 'hallucination' is:	C) Fluent output that is factually incorrect or fabricated	C) Fluent output that is factually incorrect or fabricated	✓	Mitigations include RAG grounding, citations and lower temperature.
48	Which model family uses a decoder-only architecture?	B) GPT	B) GPT	✓	GPT models generate text autoregressively with a decoder-only transformer.
49	BERT is primarily:	C) A recurrent neural network	B) An encoder-only model pretrained with masked language modeling	✗	BERT excels at understanding tasks like classification and NER.
50	Fine-tuning an LLM means:	D) Further training a pretrained model on task-specific data	D) Further training a pretrained model on task-specific data	✓	Fine-tuning adapts behavior, style or domain knowledge.
51	LoRA (Low-Rank Adaptation) is:	D) A parameter-efficient fine-tuning method that trains small low-rank adapter matrices while freezing base weights	D) A parameter-efficient fine-tuning method that trains small low-rank adapter matrices while freezing base weights	✓	LoRA cuts memory and compute; QLoRA combines it with 4-bit quantization.

#	Question	Your answer	Correct answer		Explanation
52	Model quantization means:	D) Reducing the numerical precision of weights (e.g., FP16 to INT4) to save memory and speed up inference	D) Reducing the numerical precision of weights (e.g., FP16 to INT4) to save memory and speed up inference	✓	Quantized models (like GGUF Q4) run on laptops with small quality loss.
53	RLHF stands for:	A) Reinforcement Learning from Human Feedback	A) Reinforcement Learning from Human Feedback	✓	Humans rank outputs, a reward model learns preferences, and the LLM is optimized against it.
54	In LLM applications, an 'embedding' is:	B) A numeric vector capturing the semantic meaning of text	B) A numeric vector capturing the semantic meaning of text	✓	Similar meanings produce nearby vectors, enabling semantic search.
55	Cosine similarity measures:	B) The similarity of two vectors based on the angle between them	B) The similarity of two vectors based on the angle between them	✓	Values near 1 mean very similar direction/meaning.
56	Which of these is an open-weight model you can download and run yourself?	B) Meta Llama	B) Meta Llama	✓	Open-weight models (Llama, Mistral, Gemma, Qwen) can be run locally, e.g., with Ollama.
57	The max_tokens parameter:	A) Limits the length of the generated output	A) Limits the length of the generated output	✓	Output stops once the token limit is reached.
58	A system prompt is:	B) Instructions that set the model's role and behavior for the conversation	B) Instructions that set the model's role and behavior for the conversation	✓	E.g., 'You are a helpful assistant that answers only in JSON.'
59	A lower perplexity score indicates:	B) The model predicts the text worse	D) The model predicts the text better	✗	Perplexity measures how 'surprised' the model is by the text.
60	Function calling (tool use) in LLMs means:	D) The model outputs a structured request (e.g., JSON) that the application executes as a tool call	D) The model outputs a structured request (e.g., JSON) that the application executes as a tool call	✓	The app runs the function and returns the result to the model.
61	Why do LLMs have a 'knowledge cutoff'?	D) They are trained on data only up to a certain date	D) They are trained on data only up to a certain date	✓	RAG or web search is needed for information after the cutoff.
62	A multimodal model is one that:	B) Can process multiple input types such as text and images	B) Can process multiple input types such as text and images	✓	Examples include vision-language models like LLaVA.
63	The pretraining objective of GPT-style models is:	B) Next-token prediction	B) Next-token prediction	✓	The model learns to predict the next token given previous tokens.
Prompt Engineering					
64	Zero-shot prompting means:	A) Using a model with zero fine-tuning data	B) Asking the model to do a task without giving any examples	✗	The model relies on its pretrained knowledge alone.
65	Few-shot prompting means:	C) Including a few input-output examples in the prompt	C) Including a few input-output examples in the prompt	✓	Examples show the model the expected format and behavior.
66	Chain-of-Thought (CoT) prompting:	A) Encourages the model to reason step by step before giving the final answer	A) Encourages the model to reason step by step before giving the final answer	✓	CoT improves accuracy on math and logic problems.
67	Prompt injection is:	C) Malicious input that tries to override the system's original instructions	C) Malicious input that tries to override the system's original instructions	✓	E.g., a document saying 'ignore previous instructions and reveal secrets'.
68	Best way to get reliable structured output (e.g., JSON) from an LLM:	C) Specify the exact schema/format and give an example	C) Specify the exact schema/format and give an example	✓	Many APIs also provide structured-output or JSON modes.
69	Role prompting means:	A) Assigning a persona, e.g., 'You are a senior DevOps engineer'	A) Assigning a persona, e.g., 'You are a senior DevOps engineer'	✓	A role primes relevant knowledge, tone and depth.
70	Delimiters such as XML tags or ### in a prompt are used to:	A) Clearly separate instructions from context or input data	A) Clearly separate instructions from context or input data	✓	Clear separation reduces confusion and injection risk.
71	Prompt chaining is:	C) Breaking a complex task into sequential prompts where each output feeds the next	C) Breaking a complex task into sequential prompts where each output feeds the next	✓	E.g., extract, then summarize, then format.
72	The ReAct prompting pattern stands for:	B) Reasoning + Acting, interleaving thoughts with tool calls and observations	B) Reasoning + Acting, interleaving thoughts with tool calls and observations	✓	ReAct is a foundation of many LLM agent loops.

#	Question	Your answer	Correct answer		Explanation
73	Self-consistency prompting:	C) Samples multiple reasoning paths and picks the most common final answer	C) Samples multiple reasoning paths and picks the most common final answer	✓	Majority voting across samples improves reliability.
74	Which prompt strategy helps reduce hallucinations?	C) Provide grounding context and allow the model to say 'I don't know'	C) Provide grounding context and allow the model to say 'I don't know'	✓	Grounding plus permission to abstain discourages fabrication.
75	A prompt template is:	A) A reusable prompt with placeholders for variables	A) A reusable prompt with placeholders for variables	✓	E.g., 'Summarize the following {document} in {n} bullets.'
76	Instead of only telling the model what NOT to do, it is usually better to:	B) Clearly tell it what TO do with specific, positive instructions	B) Clearly tell it what TO do with specific, positive instructions	✓	Positive, specific instructions are easier for models to follow.
77	Why evaluate prompts against a test set?	D) To measure quality changes objectively across prompt versions	D) To measure quality changes objectively across prompt versions	✓	Prompt changes can fix one case while breaking others.
78	Tree of Thoughts (ToT) prompting:	B) Explores and evaluates multiple reasoning branches before choosing	B) Explores and evaluates multiple reasoning branches before choosing	✓	It generalizes CoT into a search over possible thoughts.
Retrieval-Augmented Generation (RAG)					
79	RAG stands for:	D) Retrieval-Augmented Generation	D) Retrieval-Augmented Generation	✓	RAG retrieves relevant documents and passes them to the LLM as context.
80	The main benefit of RAG is:	A) Grounding answers in external, up-to-date data without retraining the model	A) Grounding answers in external, up-to-date data without retraining the model	✓	RAG also enables citations and private-data Q&A.
81	The correct order of a typical RAG pipeline is:	B) Load, chunk, embed, store in vector DB, retrieve, generate	B) Load, chunk, embed, store in vector DB, retrieve, generate	✓	Indexing happens offline; retrieval and generation happen per query.
82	Why do we chunk documents in RAG?	D) To create smaller pieces that fit the context window and improve retrieval precision	D) To create smaller pieces that fit the context window and improve retrieval precision	✓	Chunk size is a key tuning parameter.
83	What is the purpose of chunk overlap?	C) To preserve context across chunk boundaries	C) To preserve context across chunk boundaries	✓	Overlap prevents sentences from being cut off from their context.
84	Which of these is a vector database?	D) Chroma	D) Chroma	✓	Others include FAISS, Pinecone, Qdrant, Weaviate, Milvus and pgvector.
85	FAISS is:	D) A library from Meta for efficient similarity search over dense vectors	D) A library from Meta for efficient similarity search over dense vectors	✓	FAISS supports exact and approximate nearest-neighbor indexes.
86	Top-k retrieval means:	B) Returning the k most similar chunks to the query	B) Returning the k most similar chunks to the query	✓	k trades recall against context size and noise.
87	Hybrid search combines:	A) SQL and NoSQL databases	B) Keyword search (e.g., BM25) with semantic vector search	✗	Keywords catch exact terms; vectors catch meaning.
88	BM25 is a:	B) Keyword/lexical ranking function	B) Keyword/lexical ranking function	✓	BM25 scores documents by term frequency and inverse document frequency.
89	A re-ranker in RAG:	D) Re-scores retrieved candidates (e.g., with a cross-encoder) for better relevance ordering	D) Re-scores retrieved candidates (e.g., with a cross-encoder) for better relevance ordering	✓	Retrieve broadly, then re-rank to the best few.
90	For embeddings in a RAG system you should:	C) Use the same embedding model for indexing and querying	C) Use the same embedding model for indexing and querying	✓	Vectors from different models live in different spaces and cannot be compared.
91	Metadata filtering in retrieval allows you to:	D) Restrict results using attributes like date, source or department	D) Restrict results using attributes like date, source or department	✓	E.g., only search HR documents from 2025.
92	Your RAG system keeps retrieving irrelevant chunks. A good first fix is:	C) Tune chunking and retrieval (chunk size, hybrid search, re-ranking)	C) Tune chunking and retrieval (chunk size, hybrid search, re-ranking)	✓	Most RAG quality problems are retrieval problems.
93	RAGAS is used for:	A) Evaluating RAG pipelines with metrics like faithfulness and context precision	A) Evaluating RAG pipelines with metrics like faithfulness and context precision	✓	It also measures answer relevancy and context recall.

#	Question	Your answer	Correct answer		Explanation
94	The 'faithfulness' metric in RAG evaluation checks:	A) Whether the user liked the answer	D) Whether the answer is supported by the retrieved context	✗	Low faithfulness indicates hallucination beyond the context.
95	The 'lost in the middle' problem means:	A) LLMs may underuse information placed in the middle of long contexts	A) LLMs may underuse information placed in the middle of long contexts	✓	Place the most relevant chunks at the start or end of the context.
96	If your knowledge base changes daily, the better approach is:	B) RAG	B) RAG	✓	RAG updates knowledge by re-indexing documents, not retraining.
97	Approximate Nearest Neighbor (ANN) indexes like HNSW:	A) Trade a little accuracy for much faster vector search	A) Trade a little accuracy for much faster vector search	✓	HNSW builds a navigable graph for sub-linear search.
98	LangChain and LlamaIndex are:	A) Frameworks for building LLM applications (loaders, splitters, retrievers, chains)	A) Frameworks for building LLM applications (loaders, splitters, retrievers, chains)	✓	They speed up building RAG and agent pipelines.
Advanced & Multi-RAG					
99	Multimodal RAG refers to:	D) Retrieval over multiple data types such as text, images and tables	D) Retrieval over multiple data types such as text, images and tables	✓	It often uses vision models or captions for images and tables.
100	Multi-query retrieval:	D) Uses an LLM to generate several reformulations of the query to improve recall	D) Uses an LLM to generate several reformulations of the query to improve recall	✓	Results from all variants are merged and deduplicated.
101	HyDE (Hypothetical Document Embeddings):	B) Generates a hypothetical answer, embeds it, and uses it to retrieve real documents	B) Generates a hypothetical answer, embeds it, and uses it to retrieve real documents	✓	A fake answer is often closer in vector space to real answers than the question is.
102	Agentic RAG means:	A) An agent decides when, where and how to retrieve, possibly iteratively using tools	A) An agent decides when, where and how to retrieve, possibly iteratively using tools	✓	The agent can rewrite queries, pick sources and verify answers.
103	In a multi-source RAG system, 'routing' means:	D) Directing the query to the most relevant data source or index	D) Directing the query to the most relevant data source or index	✓	E.g., HR questions go to the HR index, code questions to the code index.
104	GraphRAG uses:	B) A knowledge graph of entities and relationships to support multi-hop reasoning	B) A knowledge graph of entities and relationships to support multi-hop reasoning	✓	Graphs help answer questions that connect facts across documents.
105	Parent-document retrieval:	D) Matches on small chunks but returns the larger parent section as context	D) Matches on small chunks but returns the larger parent section as context	✓	Small chunks match precisely; large parents give full context.
106	Corrective RAG (CRAG):	D) Grades retrieved documents and, if they are poor, takes corrective action such as a web search	D) Grades retrieved documents and, if they are poor, takes corrective action such as a web search	✓	It adds a self-check step on retrieval quality.
107	Reciprocal Rank Fusion (RRF) is used to:	A) Merge ranked results from multiple retrievers into one list	A) Merge ranked results from multiple retrievers into one list	✓	Common in hybrid search to combine BM25 and vector results.
108	A 'multi-hop' question in RAG is one that:	D) Requires combining information from multiple documents to answer	D) Requires combining information from multiple documents to answer	✓	E.g., 'Who manages the team that built product X?'
Harness Engineering & AI Agents					
109	In AI, 'harness engineering' refers to:	A) Designing the scaffolding around an LLM (tools, context management, control loop, guardrails, evals) to make it a reliable agent	A) Designing the scaffolding around an LLM (tools, context management, control loop, guardrails, evals) to make it a reliable agent	✓	The same model can perform very differently depending on its harness.
110	The core loop of an LLM agent is:	C) Reason, call a tool, observe the result, and repeat until the task is done	C) Reason, call a tool, observe the result, and repeat until the task is done	✓	This loop is how agents complete multi-step tasks.
111	Context management in long-running agents involves:	C) Summarizing or compacting history and selecting relevant information to stay within the context window	C) Summarizing or compacting history and selecting relevant information to stay within the context window	✓	Good context management prevents agents from losing track or running out of tokens.

#	Question	Your answer	Correct answer		Explanation
112	Guardrails in an AI agent system are:	D) Checks on inputs, outputs and actions that keep the agent safe and on-policy	D) Checks on inputs, outputs and actions that keep the agent safe and on-policy	✓	Examples: blocking PII leaks, validating tool arguments, content filters.
113	Human-in-the-loop for agents means:	C) Requiring human approval before high-risk actions	C) Requiring human approval before high-risk actions	✓	E.g., confirming before sending emails, payments or deleting data.
114	An evaluation harness (e.g., lm-evaluation-harness) is:	D) A framework that runs models or agents against benchmark tasks and scores them automatically	D) A framework that runs models or agents against benchmark tasks and scores them automatically	✓	It makes performance comparisons repeatable.
115	Why set a maximum number of steps/iterations for an agent?	A) To prevent infinite loops and runaway cost	A) To prevent infinite loops and runaway cost	✓	Budgets on steps, time and tokens are standard harness controls.
116	Why do tool names, descriptions and schemas matter so much in agents?	C) The model decides which tool to use and how to call it based on them	C) The model decides which tool to use and how to call it based on them	✓	Clear, specific tool descriptions greatly improve tool-use accuracy.
117	In multi-agent orchestration, a supervisor/orchestrator agent:	B) Delegates subtasks to specialized agents and combines results	B) Delegates subtasks to specialized agents and combines results	✓	Frameworks like LangGraph and CrewAI support this pattern.
118	Tools like LangSmith or Langfuse provide agent observability by:	A) Tracing each step, tool call, token usage and latency for debugging	A) Tracing each step, tool call, token usage and latency for debugging	✓	Tracing is essential for debugging non-deterministic agent behavior.
Model Context Protocol (MCP) & MCP Inspector					
119	MCP stands for:	D) Model Context Protocol	D) Model Context Protocol	✓	MCP standardizes how AI apps connect to tools and data.
120	Which company introduced MCP as an open standard?	A) Google	B) Anthropic	✗	Anthropic open-sourced MCP in November 2024; it is now widely adopted.
121	The main purpose of MCP is to:	A) Replace REST APIs on the web	B) Provide an open standard for connecting AI applications to external tools and data sources	✗	Often described as 'USB-C for AI apps'.
122	The three core primitives an MCP server can expose are:	A) Tools, Resources and Prompts	A) Tools, Resources and Prompts	✓	Tools = actions, Resources = data, Prompts = templates.
123	In MCP, 'Tools' are:	A) Functions the model can invoke to perform actions	A) Functions the model can invoke to perform actions	✓	E.g., create_ticket, query_database, send_email.
124	In MCP, 'Resources' are:	A) Docker images	C) Data or content (files, records) exposed as context, identified by URIs	✗	Resources are application-controlled context, like a file's contents.
125	In MCP, 'Prompts' are:	A) Reusable prompt templates/workflows exposed by the server, typically chosen by the user	A) Reusable prompt templates/workflows exposed by the server, typically chosen by the user	✓	They often appear as slash commands in clients.
126	MCP messages use which format?	A) JSON-RPC 2.0	A) JSON-RPC 2.0	✓	Requests, responses and notifications follow JSON-RPC 2.0.
127	The standard MCP transports are:	A) Only gRPC	C) stdio and Streamable HTTP	✗	Streamable HTTP replaced the older HTTP+SSE transport.
128	The stdio transport is typically used when:	A) Streaming video	B) The MCP server runs as a local subprocess of the client	✗	The client communicates over the process's stdin/stdout.
129	In MCP architecture, each MCP client maintains:	A) A 1:1 connection with a single MCP server	A) A 1:1 connection with a single MCP server	✓	The host app (e.g., Claude Desktop or an IDE) creates one client per server.
130	The MCP Inspector is:	A) A Kubernetes monitoring dashboard	D) An interactive developer tool for testing and debugging MCP servers	✗	You can list and call tools, read resources and test prompts.
131	How do you typically launch the MCP Inspector?	A) docker compose inspect	D) npx @modelcontextprotocol/inspector	✗	Append the server's start command, e.g., npx @modelcontextprotocol/inspector python server.py
132	Which helper in the official Python SDK makes building MCP servers quick?	A) Django REST Framework	C) FastMCP	✗	from mcp.server.fastmcp import FastMCP

#	Question	Your answer	Correct answer		Explanation
133	In FastMCP, how do you register a Python function as a tool?	A) Add it to requirements.txt	D) Decorate it with @mcp.tool()	✗	Type hints and the docstring become the tool's schema and description.
134	Capability negotiation between MCP client and server happens during:	A) Every tool call	B) The initialization handshake (initialize request)	✗	Both sides declare supported features and protocol version.
135	MCP 'sampling' allows:	A) Data sampling for training	B) A server to request an LLM completion through the client	✗	This lets servers use AI without their own model API keys.
136	Which is a security best practice for MCP servers?	A) Run every server with full root access	D) Least privilege, input validation and user consent for sensitive tool calls	✗	MCP servers can take real actions, so treat them like any privileged integration.
137	How does MCP relate to LLM function calling?	A) Function calling is the model-level ability to request a tool; MCP standardizes how tools are discovered and connected across apps	A) Function calling is the model-level ability to request a tool; MCP standardizes how tools are discovered and connected across apps	✓	MCP builds on function calling to make tools reusable across hosts.
138	'Tool poisoning' in MCP refers to:	A) Corrupting the GPU drivers	C) Malicious instructions hidden in tool descriptions to manipulate the model	✗	Only install MCP servers from trusted sources and review their tools.
Ollama & Local LLMs					
139	Ollama is a tool for:	A) Running open LLMs locally on your machine	A) Running open LLMs locally on your machine	✓	It packages model weights, config and a local API server.
140	Which command downloads a model without starting a chat?	A) ollama fetch llama3	B) ollama pull llama3	✗	ollama pull downloads the model to your local store.
141	Which command starts an interactive chat with a model?	A) ollama start llama3	C) ollama run llama3	✗	ollama run also pulls the model if it is not present.
142	Which command lists models downloaded locally?	A) ollama list	A) ollama list	✓	It shows model names, sizes and modification times.
143	Ollama's local API server listens on which default port?	A) 8501	B) 11434	✗	Base URL: http://localhost:11434
144	An Ollama Modelfile is used to:	A) Build a Docker image	B) Customize a model with a base model (FROM), parameters and a system prompt	✗	Similar in spirit to a Dockerfile, but for models.
145	How do you build a custom model from a Modelfile?	A) ollama compile mymodel	C) ollama create mymodel -f Modelfile	✗	Then run it with ollama run mymodel.
146	Which Ollama REST endpoint is used for multi-turn chat with message history?	A) /api/deploy	D) /api/chat	✗	/api/generate handles single prompt completions.
147	Models served by Ollama typically use which file format?	A) TFRecord	B) GGUF (quantized)	✗	GGUF is the llama.cpp format supporting quantized weights.
148	A key advantage of running LLMs locally with Ollama is:	A) Unlimited context window	B) Data privacy and no per-token API cost	✗	Data never leaves your machine.
149	To create embeddings for RAG with Ollama you would:	A) Use the Modelfile SYSTEM line	C) Use an embedding model (e.g., nomic-embed-text) via the embed API	✗	Embedding models output vectors, not text.
150	Which command shows which models are currently loaded in memory?	A) ollama top	C) ollama ps	✗	Useful to check GPU/CPU usage of loaded models.
MLOps & MLflow					
151	MLOps is best described as:	A) Practices to deploy, monitor and maintain ML models in production reliably	A) Practices to deploy, monitor and maintain ML models in production reliably	✓	It applies DevOps ideas (CI/CD, versioning, monitoring) to ML.
152	Which are core components of MLflow?	A) Images, Layers, Volumes and Networks	D) Tracking, Projects, Models and Model Registry	✗	Tracking logs runs; the Registry manages model versions.
153	MLflow Tracking records:	A) Only Docker images	C) Parameters, metrics, artifacts and code versions of runs	✗	This makes experiments comparable and reproducible.
154	How do you log a hyperparameter in MLflow?	A) mlflow.log_param('lr', 0.01)	A) mlflow.log_param('lr', 0.01)	✓	Use log_params() to log a dict of many parameters.
155	How do you log a metric such as accuracy in MLflow?	A) mlflow.log_param('accuracy', 0.92)	C) mlflow.log_metric('accuracy', 0.92)	✗	Metrics can be logged per step to plot training curves.

#	Question	Your answer	Correct answer		Explanation
156	Which statement starts a new MLflow run?	A) mlflow.open()	D) with mlflow.start_run():	✗	Everything logged inside the block belongs to that run.
157	mlflow.autolog() does what?	A) Automatically deploys to Kubernetes	B) Automatically logs parameters, metrics and models for supported libraries	✗	Works with scikit-learn, XGBoost, PyTorch, TensorFlow and more.
158	How do you open the MLflow tracking UI locally?	A) mlflow ui (default port 5000)	A) mlflow ui (default port 5000)	✓	Then browse to http://localhost:5000
159	The MLflow Model Registry is used to:	A) Centrally version models and manage their lifecycle (e.g., aliases like 'champion')	A) Centrally version models and manage their lifecycle (e.g., aliases like 'champion')	✓	It records lineage from model version back to the training run.
160	Data drift means:	A) Slow database queries	B) The distribution of production input data changes from the training data	✗	Drift can silently degrade model accuracy.
161	Concept drift means:	A) The server's clock drifts	B) The relationship between inputs and the target changes over time	✗	E.g., customer buying behavior changing after a major event.
162	The main reason for experiment tracking is to:	A) Compare runs and reproduce results	A) Compare runs and reproduce results	✓	Without tracking, it is hard to know which settings produced the best model.
163	DVC (Data Version Control) is used to:	A) Manage Docker containers	C) Version datasets and models alongside Git	✗	Large files are stored remotely and referenced by small Git-tracked files.
164	A feature store (e.g., Feast) provides:	A) A marketplace to buy ML features	B) A central place to store and serve features consistently for training and inference	✗	It helps prevent training-serving skew.
165	Training-serving skew is:	A) A difference between how data is processed in training versus in production serving	A) A difference between how data is processed in training versus in production serving	✓	It causes models to underperform in production despite good offline metrics.
166	A canary deployment of a new model means:	A) Testing only on synthetic data	D) Routing a small percentage of traffic to the new model first	✗	If metrics look good, traffic is gradually increased.
167	A/B testing of ML models involves:	A) Comparing two Docker images by size	B) Splitting live traffic between two models and comparing business/quality metrics	✗	It measures real-world impact, not just offline accuracy.
168	Production model monitoring should track:	A) Only disk space	C) Prediction quality, latency and data drift	✗	Model health is more than infrastructure health.
169	An MLflow model 'flavor' is:	A) A dataset type	B) A convention (e.g., sklearn, pytorch) describing how to save, load and serve a model	✗	The pyfunc flavor offers a generic predict interface.
170	How can you serve version 1 of a registered MLflow model as a REST API?	A) mlflow models serve -m 'models:/my_model/1'	A) mlflow models serve -m 'models:/my_model/1'	✓	This starts a local scoring server with an /invocations endpoint.
Streamlit					
171	Streamlit is:	A) A vector database	B) A Python framework for building data and ML web apps quickly	✗	Write Python scripts; Streamlit renders the UI.
172	How do you run a Streamlit app?	A) streamlit run app.py	A) streamlit run app.py	✓	The app opens in your browser.
173	What is Streamlit's default port?	A) 11434	C) 8501	✗	Change it with --server.port.
174	Which function creates a single-line text input box?	A) st.text_input()	A) st.text_input()	✓	st.text_area() is used for multi-line input.
175	How does Streamlit's execution model work?	A) It compiles to JavaScript	C) The script reruns from top to bottom on every widget interaction	✗	This is why caching and session state are important.
176	To keep values (like chat history) across reruns, use:	A) st.cache_data	C) st.session_state	✗	session_state persists per user session.
177	To load an ML model or LLM client once and reuse it across reruns, use:	A) @st.cache_data	C) @st.cache_resource	✗	cache_resource is for unserializable shared objects; cache_data is for data like DataFrames.

#	Question	Your answer	Correct answer		Explanation
178	Which elements are used to build a chat interface in Streamlit?	A) st.chat_message and st.chat_input	A) st.chat_message and st.chat_input	✓	These make LLM chatbot UIs quick to build.
179	How do users upload a PDF for a RAG app in Streamlit?	A) st.upload()	C) st.file_uploader()	✗	You can restrict file types with type=['pdf'].
180	How do you place widgets in a side panel?	A) st.menu	B) st.sidebar	✗	E.g., st.sidebar.selectbox('Model', [...]).
Docker					
181	The main difference between containers and virtual machines is:	A) There is no difference	B) Containers share the host OS kernel and are lighter; VMs run a full guest OS	✗	This makes containers start faster and use less memory.
182	A Dockerfile is:	A) A text file with instructions to build a Docker image	A) A text file with instructions to build a Docker image	✓	Instructions include FROM, COPY, RUN and CMD.
183	Which command builds an image named myapp from the current directory?	A) docker run -t myapp .	C) docker build -t myapp .	✗	The final dot is the build context.
184	What does docker run -p 8000:80 myapp do?	A) Maps container port 8000 to host port 80	D) Maps host port 8000 to container port 80	✗	Format is -p HOST:CONTAINER.
185	Which command lists running containers?	A) docker status	C) docker ps	✗	Use docker ps -a to include stopped containers.
186	What is the relationship between an image and a container?	A) Images run, containers are stored	B) An image is a read-only template; a container is a running instance of it	✗	Many containers can run from one image.
187	The CMD instruction in a Dockerfile:	A) Sets the base image	C) Sets the default command run when the container starts (can be overridden)	✗	RUN executes at build time; CMD at container start.
188	The COPY instruction:	A) Copies environment variables	C) Copies files from the build context into the image	✗	E.g., COPY requirements.txt .
189	Docker Compose is used to:	A) Replace Kubernetes in production clusters	B) Define and run multi-container applications with a YAML file	✗	E.g., an API, a vector DB and a UI started with docker compose up.
190	Docker volumes are used to:	A) Expose ports	B) Persist data beyond the container's lifecycle	✗	Data in the container's writable layer is lost when the container is removed.
191	A common way to reduce Docker image size is:	A) Using multi-stage builds and slim base images	A) Using multi-stage builds and slim base images	✓	Build in one stage; copy only the artifacts into a small final image.
192	A .dockerignore file is used to:	A) Exclude files from the build context	A) Exclude files from the build context	✓	Exclude .git, venv, data and secrets to make builds faster and safer.
193	Docker Hub is:	A) A public registry for storing and sharing container images	A) A public registry for storing and sharing container images	✓	Push with docker push username/image:tag.
194	To make the best use of Docker layer caching, you should:	A) Disable caching always	B) Copy and install dependencies before copying frequently-changing source code	✗	Then code changes don't force reinstalling dependencies.
195	Which command shows a container's logs?	A) docker output <container>	C) docker logs <container>	✗	Add -f to follow logs live.
Kubernetes (K8s)					
196	The smallest deployable unit in Kubernetes is a:	A) Pod	A) Pod	✓	A Pod wraps one or more containers sharing network and storage.
197	A Kubernetes Deployment:	A) Exposes pods to the internet directly	C) Manages ReplicaSets to keep a desired number of pod replicas and handle rolling updates	✗	It provides declarative updates for pods.
198	A Kubernetes Service provides:	A) A stable network endpoint to access a set of pods	A) A stable network endpoint to access a set of pods	✓	Pod IPs change; the Service name/IP stays stable.

#	Question	Your answer	Correct answer		Explanation
199	Which Service type provisions an external cloud load balancer?	A) ClusterIP	C) LoadBalancer	✗	ClusterIP is internal-only; NodePort exposes a port on each node.
200	Which command creates or updates resources from a YAML file?	A) kubectl build -f deployment.yaml	C) kubectl apply -f deployment.yaml	✗	apply is declarative and idempotent.
201	Which command lists pods in the current namespace?	A) kubectl list pods	B) kubectl get pods	✗	Add -A for all namespaces.
202	Where should you store an API key used by a pod?	A) In a Kubernetes Secret	A) In a Kubernetes Secret	✓	ConfigMaps are for non-sensitive configuration.
203	Kubernetes namespaces are used for:	A) Storing secrets only	B) Logical isolation and organization of resources within a cluster	✗	E.g., separate dev, staging and prod namespaces.
204	The Horizontal Pod Autoscaler (HPA):	A) Adds more CPU to a single pod	D) Scales the number of pod replicas based on CPU or other metrics	✗	Vertical scaling (more resources per pod) is handled by VPA.
205	A Kubernetes Ingress is used to:	A) Route external HTTP(S) traffic to services based on hosts and paths	A) Route external HTTP(S) traffic to services based on hosts and paths	✓	Requires an Ingress controller such as NGINX.
206	If a pod's readiness probe fails, Kubernetes will:	A) Restart the container immediately	D) Stop sending it traffic, without restarting it	✗	A failing liveness probe is what triggers a container restart.
207	Which command shows a pod's logs?	A) kubectl describe logs	D) kubectl logs <pod-name>	✗	Add -f to stream, or -c to pick a container.
208	Which control plane component stores the cluster's state?	A) kube-proxy	B) etcd	✗	etcd is a distributed key-value store.
209	Which component assigns pods to nodes?	A) kube-scheduler	A) kube-scheduler	✓	It considers resource requests, affinity and taints.
210	How do you roll back a Deployment to its previous version?	A) kubectl rollout undo deployment/<name>	A) kubectl rollout undo deployment/<name>	✓	Check history with kubectl rollout history.
Git & GitHub					
211	Which command copies a remote repository to your machine?	A) git clone <url>	A) git clone <url>	✓	Forking is done on GitHub; cloning copies it locally.
212	What does git add do?	C) Stages changes for the next commit	C) Stages changes for the next commit	✓	Staging lets you choose what goes into a commit.
213	Which command records staged changes with a message?	C) git commit -m 'message'	C) git commit -m 'message'	✓	Commits are stored locally until pushed.
214	Which command uploads local commits to a remote repository?	D) git push	D) git push	✓	E.g., git push origin main
215	git pull is equivalent to:	C) git fetch followed by git merge	C) git fetch followed by git merge	✓	It downloads and integrates remote changes.
216	A Git branch is:	C) An independent line of development	C) An independent line of development	✓	Branches let you work on features without affecting main.
217	A Pull Request (PR) on GitHub is used to:	C) Propose merging changes and request code review	C) Propose merging changes and request code review	✓	PRs can run CI checks before merging.
218	A merge conflict happens when:	B) The same lines are changed differently in two branches being merged	B) The same lines are changed differently in two branches being merged	✓	Resolve manually, then commit.
219	The .gitignore file is used to:	A) Specify files Git should not track	A) Specify files Git should not track	✓	E.g., .env, __pycache__, venv/, large data files.
220	Which command shows modified, staged and untracked files?	D) git status	D) git status	✓	git log shows commit history instead.
221	GitHub Actions workflows are defined in:	B) YAML files under .github/workflows/	B) YAML files under .github/workflows/	✓	They run CI/CD on events like push and pull_request.
222	Where should API keys used by GitHub Actions be stored?	A) In GitHub repository Secrets	A) In GitHub repository Secrets	✓	Access them with ${{ secrets.MY_KEY }}.
APIs & FastAPI					
223	REST is:	B) An architectural style using HTTP methods to operate on resources	B) An architectural style using HTTP methods to operate on resources	✓	Resources are identified by URLs, e.g., /users/42.
224	Which HTTP method is used to retrieve data?	D) GET	D) GET	✓	GET should not change server state.
225	Which HTTP method is typically used to create a new resource or submit data?	D) POST	D) POST	✓	E.g., POST /predict with input JSON.

#	Question	Your answer	Correct answer		Explanation
226	HTTP status code 404 means:	B) Not Found	B) Not Found	✓	The requested resource does not exist.
227	HTTP status code 401 means:	B) Unauthenticated: valid credentials are missing or invalid	B) Unauthenticated: valid credentials are missing or invalid	✓	403 means authenticated but not allowed.
228	FastAPI is:	D) A high-performance Python web framework with automatic OpenAPI docs	D) A high-performance Python web framework with automatic OpenAPI docs	✓	It is popular for serving ML models.
229	Where are FastAPI's interactive Swagger docs available by default?	C) /docs	C) /docs	✓	ReDoc is available at /redoc.
230	How do you run a FastAPI app defined as app in main.py during development?	A) uvicorn main:app --reload	A) uvicorn main:app --reload	✓	--reload restarts on code changes.
231	In FastAPI, Pydantic is used for:	B) Request/response data validation using type hints	B) Request/response data validation using type hints	✓	Invalid input returns a 422 error automatically.
232	Which HTTP method is NOT idempotent?	A) POST	A) POST	✓	Repeating a POST can create duplicate resources.
233	The most common data format for REST API payloads is:	B) JSON	B) JSON	✓	Content-Type: application/json
234	HTTP status code 429 means:	A) Too Many Requests (rate limited)	A) Too Many Requests (rate limited)	✓	Common with LLM APIs; handle with retry and exponential backoff.
Deployment (Render)					
235	Render is:	C) A cloud platform (PaaS) to deploy web services, static sites and databases from Git	C) A cloud platform (PaaS) to deploy web services, static sites and databases from Git	✓	Connect a GitHub repo and Render builds and deploys it.
236	For a web service on Render, your app should listen on:	C) 0.0.0.0 and the port in the PORT environment variable	C) 0.0.0.0 and the port in the PORT environment variable	✓	Binding to localhost makes the app unreachable from outside.
237	With auto-deploy enabled on Render:	B) Pushing to the connected branch automatically triggers a new deploy	B) Pushing to the connected branch automatically triggers a new deploy	✓	This gives simple continuous deployment.
238	Where should API keys (e.g., OPENAI_API_KEY) go when deploying on Render?	D) Environment variables in the Render dashboard	D) Environment variables in the Render dashboard	✓	Never commit secrets to the repository.
239	Why is the first request to a free-tier Render service sometimes slow?	C) The instance spins down after inactivity and must cold start	C) The instance spins down after inactivity and must cold start	✓	Paid instances stay always on.
240	A suitable start command for a Streamlit app on Render is:	D) streamlit run app.py --server.port $PORT --server.address 0.0.0.0	D) streamlit run app.py --server.port $PORT --server.address 0.0.0.0	✓	Bind to Render's PORT and all interfaces.
Jenkins & CI/CD					
241	Jenkins is:	A) An open-source automation server used for CI/CD	A) An open-source automation server used for CI/CD	✓	It automates build, test and deploy steps.
242	A Jenkinsfile is:	C) A file defining the pipeline as code, stored in the repository	C) A file defining the pipeline as code, stored in the repository	✓	Pipeline-as-code makes CI versioned and reviewable.
243	In a declarative Jenkinsfile, the top-level block is:	A) pipeline { }	A) pipeline { }	✓	Inside it: agent, stages, post, etc.
244	The correct nesting in a declarative pipeline is:	C) stages > stage > steps	C) stages > stage > steps	✓	Each stage contains steps such as sh 'pytest'.
245	The best way to trigger a Jenkins build automatically on every GitHub push is:	B) A GitHub webhook	B) A GitHub webhook	✓	Webhooks notify Jenkins immediately when code changes.
246	Continuous Integration (CI) means:	A) Frequently merging code into a shared branch with automated builds and tests	A) Frequently merging code into a shared branch with automated builds and tests	✓	CI catches integration bugs early.
247	Continuous Delivery differs from Continuous Deployment because:	D) Delivery keeps code always deployable but needs manual approval for production; Deployment releases automatically	D) Delivery keeps code always deployable but needs manual approval for production; Deployment releases automatically	✓	Both rely on a strong automated pipeline.

#	Question	Your answer	Correct answer		Explanation
248	Jenkins runs on which port by default?	B) 8080	B) 8080	✓	Access it at http://localhost:8080
249	How should secrets like Docker Hub passwords be used in Jenkins pipelines?	A) Store them in Jenkins Credentials and access them with withCredentials	A) Store them in Jenkins Credentials and access them with withCredentials	✓	Jenkins masks credential values in logs.
250	In Jenkins, an 'agent' is:	D) A machine or container where pipeline steps execute	D) A machine or container where pipeline steps execute	✓	agent any runs on any available executor.